



SpiderVault 3.0

3 kW Single-Phase Solar Hybrid Inverter

SPECIFICATION SHEET

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AC INPUT	
Rated Power	3000 W
Input Voltage	230 VAC
Selectable Voltage Range	190 – 260 VAC
Frequency Range	50 Hz / 60 Hz Auto Sensing

AC OUTPUT	
AC Voltage (Battery Mode)	230 VAC ± 5%
Surge Power	6000 VA
Peak Efficiency	97%
Switching Modes	On Grid Off Grid Hybrid (Optional)
Transfer Time	10 ms
Waveform	Pure Sine Wave

SOLAR CHARGER & AC CHARGER			
Solar Charger Type	MPPT	No. of MPPTs	1
Max. PV Array Open Circuit Voltage	450 VDC	Max. PV Array Power	8500 W
Max. PV Input Current	27 A	MPP Voltage Range	60 – 450 VDC
Maximum Solar Charge Current	100 A	Maximum AC Charge Current	60 A
Maximum Charge Current	100 A		

BATTERY	
Battery Cell Technology	Lithium Ion
Nominal Voltage	48 V
Battery Capacity	3 kWh
Max. Continuous Charging Current	10 A
Max. Continuous Discharging Current	60 A
Full Charging Voltage	54.3 V
Quantity of Batteries	1 Nos

PROTECTION	
Protection System	5th Gen BMS DC Breaker
Anti-Islanding Protection	Integrated
AC Over Current Protection	Integrated
Output Short Circuit	Integrated
Overload Protection	Integrated
Battery Voltage Too Low	Integrated
Temperature Too High	Integrated
Input Voltage Out of Range / Ripple Too High	Integrated
Fan Block	Integrated
IP Protection	IP65

GENERAL DATA	
Dimensions L × W × H	362 × 155 × 728 mm
Net Weight	34 kg
Installation Style	Wall-Mounted
Topology	High Frequency Isolation
Operating Temperature	-10°C ~ 60°C
Cooling	Advanced Nano PCM + Forced Convection
Display	LCD with indicators (mode / load / input / output)
PV Connection	MC4 Connector
AC Connection	Terminal Connectors
No. of AC Outputs	1
Noise Level	< 50 dB
Communication Interface	BMS Dry-Contact CAN Bluetooth Wi-Fi AI Cloud
Warranty	5 Years · Extended 12 Years

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CERTIFICATIONS & STANDARDS

BIS CERTIFICATION

- BIS Certified (R-63005436)

SAFETY & GRID

- IS 16221 (Part 2): 2015 / IEC 62109-2: 2011
- IS 16169: 2019 / IEC 62116: 2014
- IS 17980: 2022 / IEC 62891: 2020

3 kW

Rated Power

97%

Peak Efficiency

3 kWh

Battery Capacity

IP65

IP Protection